

Sunish Kumar

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Education

Institute of Engineering and Management

B.Tech in Electronics and Communication Engineering

Aug 2019 – June 2023

CGPA: 8.8

Experience

PwC India, Kolkata

Aug 2023 – Present

Software Engineer – Full Stack

- Developed scalable backend services using **Java, Spring Boot, Microservices and REST APIs**, designing modular service-layer architectures for enterprise applications.
- Designed and optimized **15+ REST APIs**, improving average response time by **25%** through query optimization, efficient data access and backend refactoring.
- Implemented **WebSocket-based real-time communication**, reducing update latency by **40%** for interactive application workflows.
- Designed **MongoDB schemas and indexes** for applications handling **10K+ records**, improving query performance for frequently accessed data.
- Implemented secure authentication and authorization using **JWT, AWS Cognito and Azure AD/SSO**, including role-based access control for backend APIs.
- Built backend-integrated applications using **React.js, Vue.js, Next.js and Redux**; worked with AWS/Azure services and YAML-based CI/CD pipelines.

Projects

MemoryGraph – Private Personal Memory Platform

Live GitHub

- Designed and built a **modular monolith** using **Java, Spring Boot, PostgreSQL and JPA**, with separated API, application, domain and infrastructure layers.
- Designed REST APIs for **authentication, memories, media, search, imports, people, places, trips and question answering**, with centralized validation and exception handling.
- Implemented **hybrid search** using PostgreSQL full-text search and **pgvector HNSW** vector search, combining keyword and semantic retrieval using Reciprocal Rank Fusion.
- Built an **asynchronous processing pipeline** with persistent jobs, bounded worker pools, retries and exponential backoff for metadata extraction, OCR, captions and embeddings.
- Implemented concurrent job processing using **SELECT FOR UPDATE SKIP LOCKED**, preventing duplicate job claims across multiple application instances.
- Implemented **JWT authentication, owner-scoped authorization and deny-by-default security**; added Flyway migrations, Docker Compose, Testcontainers and structured request correlation.

Technical Skills

Languages: Java, JavaScript (ES6+), TypeScript, SQL

Backend: Spring Boot, Spring MVC, Spring Data JPA, Hibernate, REST APIs, Microservices, JWT, WebSocket, Maven

Databases: PostgreSQL, MongoDB, MySQL, pgvector, Indexing, Query Optimization, Transactions

System Design: HLD, LLD, Modular Monolith, Microservices, Caching, Load Balancing, Concurrency, Async Processing

Cloud/DevOps: AWS Amplify, AWS Cognito, Azure Static Web Apps, Azure AD, Azure CLI, Docker, Docker Compose, Railway, CI/CD

Testing/Tools: JUnit, Testcontainers, Postman, Git, GitHub, Flyway

Frontend: React.js, Next.js, Vue.js, Redux, Tailwind CSS, Material-UI

Achievements

• **TCS CodeVita:** Global Rank 267

Google Hash Code 2022: Global Rank 4887